

A physics-aware recurrent surrogate for nonlinear viscoelastic tendon force prediction

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Problem Definition

This project studies force prediction for a tendon-like strip undergoing ramp-hold uniaxial extension. The underlying FEBio model uses a tendon elastic response wrapped by an uncoupled viscoelastic material, so the effective constitutive behavior is nonlinear and time dependent (FEBio 2025). Rather than predicting full finite-element fields, the surrogate targets the global reaction force history $F(t)$, which is the quantity needed for reduced musculoskeletal simulation.

Following the FEBio Theory Manual (FEBio 2025), the generator is motivated by transient solid mechanics with additive elastic and viscous stress contributions,

$$\rho \ddot{\mathbf{u}} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{b}, \quad \boldsymbol{\sigma} = \boldsymbol{\sigma}_e + \boldsymbol{\sigma}_v,$$

but the learned surrogate ultimately approximates only the reduced global force response rather than the full displacement or stress field.

The final data-generation setup was a fixed-geometry tensile test:

- rectangular strip: 10 mm $\times$ 2 mm $\times$ 1 mm,
- loading axis: x,
- target displacement: 0.5 mm,
- target strain: 0.05,
- total time: 1.0 s,
- fixed output grid: 201 time points.

Data

Synthetic trajectories were generated in FEBio (FEBio 2025) by varying the four parameters most closely tied to the proposal goals:

- fiber stiffness: `tendon_l1` {3.0, 4.0, 5.0, 6.0}
- viscoelastic amplitude: `visco_g1` {0.30, 0.45, 0.60, 0.75}
- relaxation time: `visco_t1` {0.10, 0.20, 0.30, 0.40}
- loading rate: `ramp_time` {0.10, 0.20, 0.30, 0.40}

A $4 \times 4 \times 4 \times 4$ design produced 256 planned simulations. Of these, 240 validated successfully after processing and QC. The 16 failures were localized to the numerically difficult corner with highest stiffness and fastest loading (`tendon_l1` = 6.0, `ramp_time` = 0.1). All validated runs were resampled onto the same 201-point time grid to produce a model-ready dataset.

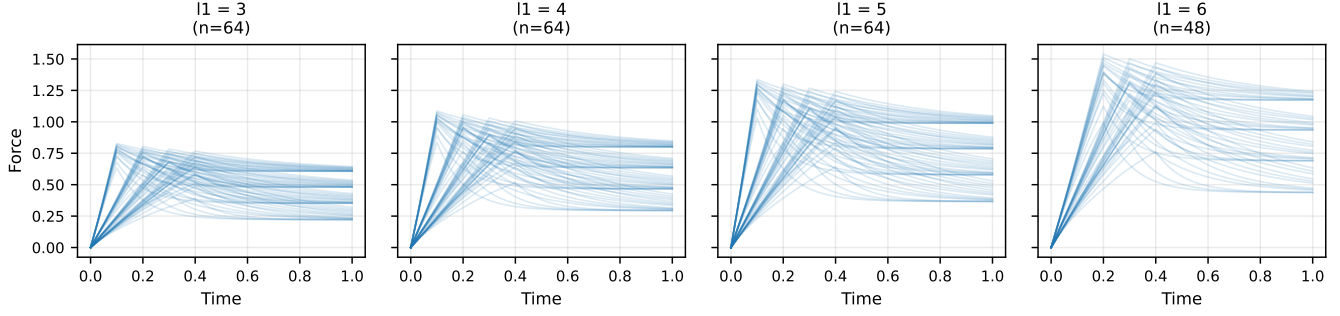


Figure 1: Validated force-time trajectories grouped by stiffness level. The systematic increase in peak and hold force with increasing tendon_l1 makes the stiffness-holdout split a meaningful extrapolation test.

Method

The learning problem is a sequence-to-sequence surrogate rather than a full field predictor. Each trajectory is represented by:

- dynamic inputs: `strain`, `strain_rate`, `phase`
- static inputs: `tendon_l1`, `visco_g1`, `visco_t1`, `ramp_time`
- target output: `force`

Two recurrent surrogate models were evaluated.

GRU baseline

The baseline uses a gated recurrent unit (GRU) sequence model (Cho et al. 2014) with:

- an MLP encoder for the static parameters,
- a GRU sequence model for the dynamic trajectory,
- a readout MLP that maps recurrent hidden states to force.

This model is flexible and accurate, but it does not explicitly enforce constitutive structure beyond recurrence.

Physics-aware coupled Maxwell model with explicit tendon scaling

The final physics-aware model is not a literal PARCv2 implementation. Instead, it is a PARC-inspired constitutive recurrent surrogate that borrows the idea of structured state evolution from PARCv2 (Nguyen et al. 2024) while using the classical generalized Maxwell picture as its viscoelastic inductive bias (Christensen 1982). In a generalized Maxwell model, an equilibrium branch carries the long-time elastic response, while several parallel Maxwell branches contribute transient overstress terms that decay with their own characteristic relaxation times during a strain hold (Christensen 1982). The model therefore structures the dynamics as a generalized Maxwell-style tendon surrogate:

- a learned elastic force scale driven by strain history,
- transient Maxwell overstress branches that build during the ramp and decay during hold,
- exact enforcement of $F(0) = 0$,
- an explicit monotone tendon stiffness scaling law applied using the raw stiffness parameter,

$$s(\text{tendon_l1}) = s_0 \left(\frac{\text{tendon_l1}}{l_{\text{ref}}} \right)^\alpha, \quad s_0 > 0, \alpha > 0.$$

This explicit scaling was introduced after simpler learned calibrators still underpredicted the force level on unseen stiffness values. It gives the model a direct, physically meaningful mechanism for increasing force with tendon stiffness. In reduced constitutive form, the final surrogate assumes

$$F(t) = s(\text{tendon_l1}) \left[g_{\text{eq}} F_{\text{full}}(\varepsilon(t); \theta_e) + \sum_i q_i(t) \right],$$

$$\frac{dq_i}{dt} = g_i d(t) - \frac{q_i}{\tau_i}, \quad d(t) = \max\left(\frac{dF_{\text{full}}}{dt}, 0\right),$$

where F_{full} is the learned elastic force scale, q_i are transient Maxwell overstress states, and τ_i are learned relaxation times. In other words, the model approximates tendon force as an equilibrium elastic contribution plus a sum of decaying nonequilibrium branches, which is the qualitative behavior expected from a generalized Maxwell material (Christensen 1982). This structure makes stress relaxation during a strain hold explicit rather than leaving it entirely to a generic recurrent hidden state.

Final physics-aware model: coupled Maxwell surrogate with explicit tendon scaling

Neural encoders map loading and material inputs to elastic, viscous, and scaling branches; explicit Maxwell memory carries the time dependence.

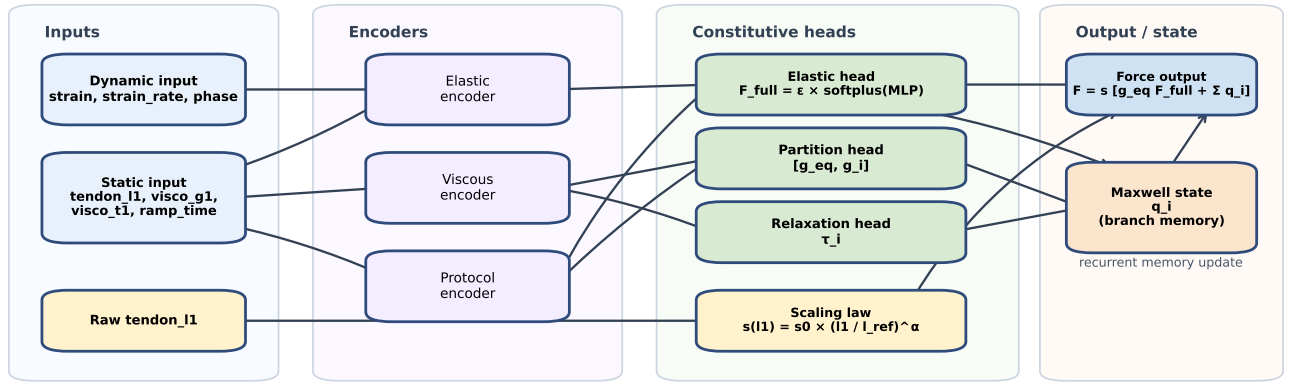


Figure 2: Architecture of the final physics-aware model. Small neural subnetworks encode static parameter groups and estimate elastic/viscous quantities, while explicit Maxwell branch states carry the viscoelastic memory over time.

Both models were trained with mean-squared error loss. Two evaluation protocols were used:

1. **Interpolation split:** mixed train/val/test assignment across the parameter grid (168 / 36 / 36 simulations).
2. **Stiffness holdout split:** whole stiffness levels held out,
 - train: `tendon_l1` {3.0, 4.0}
 - val: `tendon_l1` = 5.0
 - test: `tendon_l1` = 6.0

The second split is intentionally harder and probes extrapolation to unseen stiffness levels.

Results

Quantitative summary

Model	Evaluation setting	Train MAE	Val MAE	Test MAE	Val MSE	Test MSE
GRU baseline	Interpolation split	0.00710	0.00644	0.00818	6.85e-05	1.06e-04
Explicit Maxwell model	Interpolation split	0.00861	0.00834	0.00831	1.15e-04	1.10e-04
GRU baseline	Stiffness holdout	0.01643	0.06507	0.16790	6.33e-03	4.10e-02
Explicit Maxwell model	Stiffness holdout	0.01018	0.02466	0.04947	9.64e-04	3.21e-03

The GRU baseline remains an excellent interpolation model, but the new explicit-tendon-scaling Maxwell surrogate is now nearly tied with it on the interpolation test set (0.00831 vs. 0.00818 test MAE). The much more important result is on the stiffness-holdout benchmark: the explicit Maxwell model reduces holdout MAE from 0.0651 to 0.0247 on validation and from 0.1679 to 0.0495 on test. This is about a 71% reduction in test MAE while simultaneously improving physical plausibility.

Computational cost

Model	Parameter count	CPU ms / batch	CPU ms / simulation
GRU baseline	56,065	14.51	0.907
Explicit Maxwell model	20,748	33.26	2.079

Both surrogate models are lightweight compared with the FEBio generator. The GRU baseline has more trainable parameters because it carries a 2-layer recurrent hidden state, while the explicit Maxwell model replaces that generic recurrence with four explicit viscoelastic modes and a few small MLP heads. The timing numbers above were measured on CPU in the report execution environment and are intended as relative complexity indicators rather than a hardware-optimized benchmark.

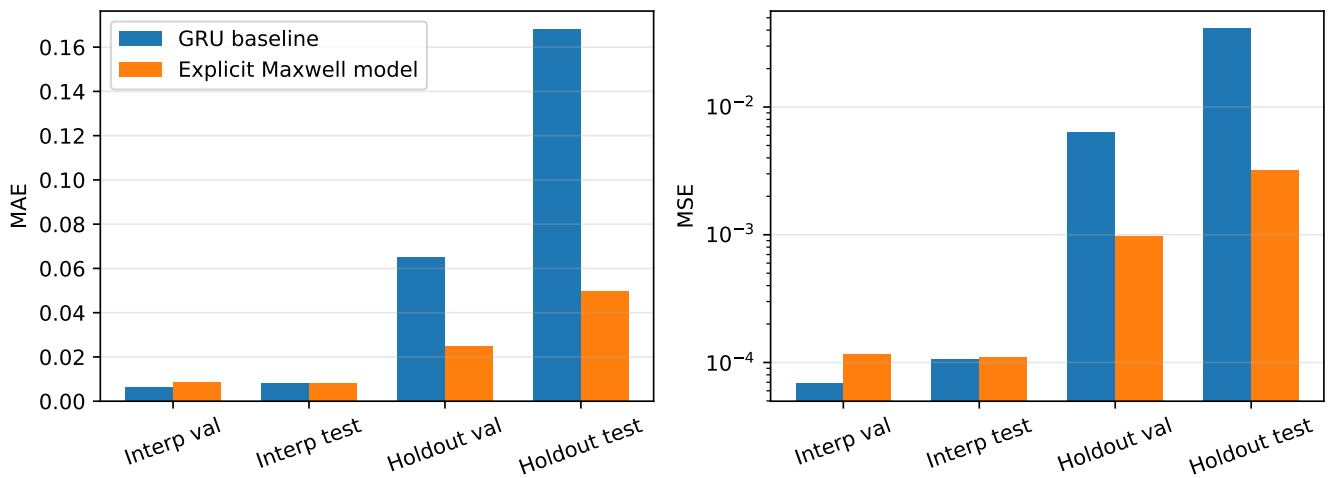


Figure 3: Comparison of the GRU baseline and the explicit-tendon-scaling Maxwell model across both evaluation protocols. The GRU is slightly better on interpolation validation, but the explicit Maxwell model dominates the stiffness-holdout benchmark.

Stiffness-holdout diagnostics

On the interpolation split, the explicit Maxwell model still captures the ramp-hold behavior well, so the stronger holdout performance is not a tradeoff against in-range accuracy. The stiffness-holdout benchmark is the clearest discriminator between the two models: the GRU baseline develops a nonphysical force offset at $t = 0$ and substantially underpredicts the force level on unseen $t_{\text{endon_l1}} = 6.0$ trajectories, while the explicit Maxwell model preserves realistic relaxation during hold, maintains exact $F(0) = 0$, and substantially reduces the force-scale bias.

Model	Split	Mean predicted $F(0)$	Mean predicted relaxation	Mean true relaxation	Mean abs. relaxation error
GRU baseline	Validation	0.2216	0.3173	0.3649	0.0580
Explicit Maxwell model	Validation	0.0000	0.3657	0.3649	0.0104
GRU baseline	Test	0.3396	0.2989	0.3390	0.0624
Explicit Maxwell model	Test	0.0000	0.3443	0.3390	0.0109

The table above highlights the main physical improvement. On the holdout test set, the GRU predicts a mean initial force of 0.3396, while the explicit Maxwell model predicts exactly 0.0. The explicit Maxwell model also matches the mean relaxation ratio much more closely (0.3443 predicted vs. 0.3390 true) than the GRU (0.2989 predicted).

Stiffness-holdout trajectory comparison

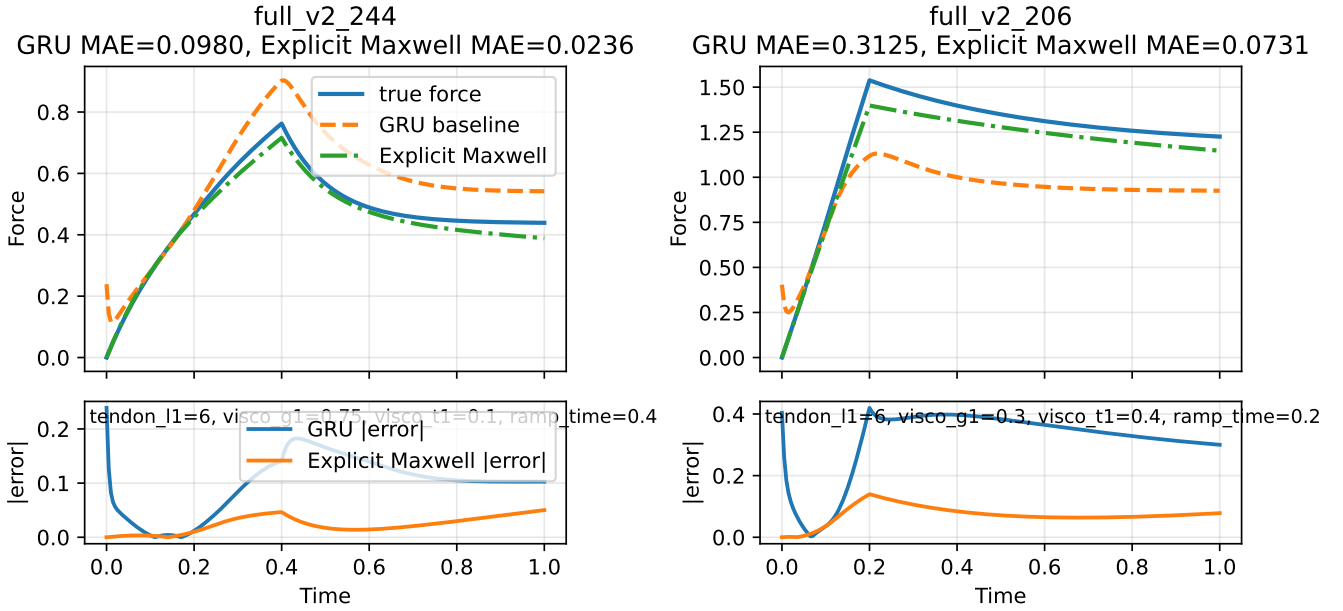


Figure 4: Direct comparison on the best and worst stiffness-holdout test cases selected from the explicit Maxwell model. Relative to the GRU baseline, the explicit Maxwell model removes the nonphysical start-force offset, tracks the ramp scale more closely, and preserves the hold-phase relaxation curve.

Even on the hardest holdout case (full_v2_206), the explicit Maxwell model lowers the error dramatically and maintains the correct qualitative shape. The remaining discrepancy is mostly a modest residual underprediction of the force level, not a breakdown of the constitutive behavior itself.

Conclusion

The FEBio pipeline successfully produced a high-quality dataset of validated tendon force trajectories, and the modeling study now gives a clear comparison between a strong generic recurrent baseline and a more physically structured constitutive surrogate.

The main findings are:

- The GRU baseline is an excellent interpolation model on this smooth, fixed-protocol sequence dataset.
- A generalized Maxwell-style recurrent model with explicit tendon stiffness scaling can nearly match the GRU on interpolation.
- On the harder stiffness-holdout benchmark, the explicit Maxwell model is substantially better than the GRU baseline.
- The explicit Maxwell model enforces $F(0) = 0$ exactly and reproduces hold-phase relaxation much more faithfully under extrapolation.
- The remaining holdout error is mainly a reduced force-level bias, not a failure to capture the viscoelastic relaxation mechanism.

Overall, the final project supports the central hypothesis that adding constitutive structure and physically meaningful parameter scaling can materially improve out-of-distribution generalization, even when a less structured recurrent model is already strong on interpolation.

Future work should focus on three directions. First, the present explicit-Maxwell surrogate could be extended toward a more faithful differentiator–integrator PARCv2-style training procedure, separating learned rate-law estimation from correction/integration. Second, the data space could be broadened by adding more stiffness levels, additional loading protocols, and possibly simple geometric variations to probe generalization beyond the current ramp-hold strip test. Third, a more explicit timing comparison against FEBio would strengthen the case for surrogate deployment in larger multiscale or musculoskeletal simulations.

References

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